

Quasivoid, Distinguishable-particle Lattice Model, and Configuration-tree Theory of Glass

by

Mysteries of Glass

The physics of glass studies materials called glass formers which can be cooled to a disordered solid state at the glass transition temperature. Examples include not only common window glass, but also glassy polymers, metallic glasses, etc. Philip Anderson, a Nobel Laureate, wrote in 1995 that **“The deepest and most interesting unsolved problem in solid state theory is probably the theory of the nature of glass and the glass transition.”** Interestingly, a related and a very important problem of spin glass may be regarded by some as just an “appetizer” before moving on to glass. Theoretical research on glass is surprisingly controversial. A famous comment by Harvard physicist David Weitz is that **“There are more theories of the glass transition than there are theorists who propose them.”** At present, not only that there is no encompassing theory of glass agreeable by most researchers, even the most fundamental questions such as whether the glass transition is thermodynamically or kinetically driven are highly controversial. Below, we briefly introduce our approach which has been showing signs of promise so far. Differences from other studies in the literature are pointed out in the footnotes.

Particle hop under deep supercooling

Motions of individual particles can be observed directly in molecular dynamics (MD) simulations of glass formers and in experiments or simulations of glassy colloidal systems of packed spheres. Supercooling is approached by decreasing the temperature or by increasing the packing fraction respectively. In moderately supercooled liquids, particles tend to move collectively with surrounding particles. By contrast, under deep supercooling, particles only vibrate about their meta-stable positions most of the time. Occasionally, a local string of a few particles hop to displace the adjacent ones, leading to a string-like particle hopping motion. A string is usually short and involves on average only two particles.¹ These hops in-

volve displacements comparable to the inter-particle separations.² They are rare activated events, but dominate the dynamics, which is slow. A signature is a secondary peak in the particle displacement probability distribution, i.e. the Van Hove correlation function. **The glass transition occurs at deep supercooling. The relevant regime thus occurs when particle hops over distances comparable to their separations dominate.** This eliminates liquid-like collective-flow characters and greatly simplifies the problem.

Trajectory of a particle, i.e. monomer, in a MD simulation of
Probability distribution of particle displacement during time

Back-and-forth particle hops

The rate of particle hops decreases as temperature decreases as in typical activated dynamics. Another important factor of slowing down is due to back-and-forth hops. For each particle hop, we observe a subsequent returning hop with a probability of up to 93% in molecular dynamics simulations (3) and 80% in colloidal experiments (11), which can be extrapolated to even higher values at deeper supercooling. **Our view is that glassy slowdown is mainly because the return probability of particle hops approaches unity, i.e. nearly all hops are reversed. A small fraction of hops which are not reversed then dominates the dynamics (3).**³

(Above) Three particles in a 3D polymer simulation which hop

Mobile layer in glassy polymer film

We first recognized the importance of repetitive particle hops
Particle hopping rate $R(\tau)$ as a function of perpendicular co

¹ Despite some researchers expecting a divergence at low temper-

ature, string lengths we have measured are in agreement with some other works which report a slight decrease as temperature decreases.

² Some works define hops or jumps over much smaller distances. Those occur at more moderate supercooling and are non-activated as they are not associated with any peak in the Van Hove correlation function.

³ Most other theories assume that the glassy slowdown is mainly because of a decrease in the particle hopping rate. We are not aware of any other theory suggesting that it is mainly because of increased back-and-forth hopping tendency.

Quasivoid

Glass formers in general consist of rather hard-core particles, the hops of which necessarily require empty spaces, i.e. free volume. Eight particles in a 3D glassy polymer simulation which hops repeatedly. Since particles cannot overlap, the white particles

Particle motions in an experiment on 2D colloidal system of micron-sized spheres exhibiting coexisting glassy (white) and crystalline (red) regions. (a) Coarse-grained particle trajectories during 8 hours of observation and (b-f) the same trajectories split up over consecutive time sub-intervals. Red dots show initial particle positions. (b) A void moves from the glass to the crystal and (d) moves back to the glass. Typical string-like motions are induced in the glassy region. Insets in (b-f): Trajectories shown together with initial particle configurations of the time subinterval. The void takes the form of a vacancy (blue solid circles) in the crystal (c-d) and a quasivoid with fragmented free volumes (blue areas) in the glass (b,e,f) (11).

Trapping and facilitated motion of quasivoids

Assuming that particle hops are induced by quasivoids, back-and-forth hops at deep supercooling then signify the trapping of quasivoids.

Motion of 6 particles from 3D polymer simulations during a pair-interaction between two strings illustrated using

Distinguishable-particle lattice model

MD simulations are widely believed to be sufficiently realistic for capturing the essence of glassy dynamics. Nevertheless, it is still a challenge to identify the relevant physical processes in MD. Lattice models are much simpler. If they manage to capture the essential physics, they can be very helpful for bridging between MD results and theories. We introduced a **distinguishable-particle lattice model (DPLM)** primarily to illustrate the facilitated motions of quasivoid we have observed in MD simulations (12). It then proves to be useful for testing and inspiring further theoretical concepts and calculations. We study a range of phenomena using the same model with no or few modifications. In each of a series of papers, we focus on how the DPLM reproduces and contributes to the explanation of a certain glassy phenomenon. However, the more important question is whether and how a single model can capture all these diverse phenomena in a consistent and unified manner. We believe that the DPLM has been making good progress and is very promising towards this aim.

In the DPLM, each particle on a square lattice is of its own type. Any two nearest neighboring particles k and l in

Videos on equilibrium particle dynamics from DPLM simulations with 1592 particles (randomly colored) and 8 vo

Random local configuration-tree theory

We construct a microscopic theory of glass in finite d

Sketch of the local configuration tree landscape for a local re

Particle diffusion coefficient D against reciprocal temperature $1/T$

Mean squared displacement (MSD) versus time t for decreasing

REFERENCES

- [1] Hai-Yao Deng, Chun-Shing Lee, Matteo Lulli, Ling-Han Zhang, and Chi-Hang Lam. Configuration-tree theoretical calculation of the mean-squared displacement of particles in glass formers. *Journal of Statistical Mechanics: Theory and Experiment*, 2019(9):094014, 2019.
- [2] Xin-Yuan Gao, Hai-Yao Deng, Chun-Shing Lee, JQ You, and Chi-Hang Lam. Emergence of two-level systems in glass formers: a kinetic monte carlo study. *arXiv preprint arXiv:2109.02275*, 2021.
- [3] Chi-Hang Lam. Repetition and pair-interaction of string-like hopping motions in glassy polymers. *The Journal of Chemical Physics*, 146(24):244906, 2017.
- [4] Chi-Hang Lam. Deeper penetration of surface effects on particle mobility than on hopping rate in glassy polymer films. *The Journal of Chemical Physics*, 149:164909, 2018.
- [5] Chi-Hang Lam. Local random configuration-tree theory for string repetition and facilitated dynamics of glass. *Journal of Statistical Mechanics: Theory and Experiment*, 2018(2):023301, 2018.
- [6] Chun-Shing Lee, Hai-Yao Deng, Cho-Tung Yip, and Chi-Hang Lam. Large heat-capacity jump in cooling-heating of fragile glass from kinetic monte carlo simulations based on a two-state picture. *Physical Review E*, 104:024131, 2021.
- [7] Chun-Shing Lee, Matteo Lulli, Ling-Han Zhang, Hai-Yao Deng, and Chi-Hang Lam. Fragile glasses associated with a dramatic drop of entropy under supercooling. *Physical Review Letters*, 125:265703, 2020.
- [8] Matteo Lulli, Chun-Shing Lee, Hai-Yao Deng, Cho-Tung Yip, and Chi-Hang Lam. Spatial heterogeneities in structural temperature cause kovacs' expansion gap paradox in aging of glasses. *Physical Review Letters*, 124:095501, 2020.
- [9] Matteo Lulli, Chun-Shing Lee, Ling-Han Zhang, Hai-Yao Deng, and Chi-Hang Lam. Kovacs effect in glass with material memory revealed in non-equilibrium particle interactions. 2021(9):093303, 2021.
- [10] Zhaohui Yang, Yoshihisa Fujii, Fuk Kay Lee, Chi-Hang Lam, and Ophelia K. C. Tsui. Glass transition dynamics and surface layer mobility in unentangled polystyrene films. *Science*, 328(5986):1676, 2010.
- [11] Cho-Tung Yip, Masaharu Isobe, Chor-Hoi Chan, Simiao Ren, Kin-Ping Wong, Qingxiao Huo, Chun-Sing Lee, Yuen-Hong Tsang, Yilong Han, and Chi-Hang Lam. Direct evidence of void-induced structural relaxations in colloidal glass formers. *Physical Review Letters*, 125:258001, 2020.
- [12] Ling-Han Zhang and Chi-Hang Lam. Emergent facilitation behavior in a distinguishable-particle lattice model of glass. *Physical Review B*, 95:184202, 2017.